

Database Foundations

3-2: Tracking Data Changes

Practices

Exercise 1: Track Data Change over Time

Overview

In this practice you use scenarios to identify data that changes over time and construct ERD models that incorporate the element of data over time.

Tasks

1. Construct the ERD for the given scenario.

In the Academic Database a Grade is issued to each STUDENT for each COURSE taken and stored in the STUDENT COURSE DETAIL entity. A STUDENT may decide to re-take a COURSE to better their Grade. The administration would like to keep a record of the old/previous Grade as well as the new Grade. Show how the ERD would be modified to include historical Grades if the STUDENT should have them. **** We will not make this actual change to the ERD.**

STUDENT COURSE DETAIL

Course ID
Student ID
* Grade

2. Examine the ERD that represents classroom assignments for different exams.
 - a. Why is start time part of the UID of ASSIGNMENT?
 - b. Name at least three time-related constraints. For example: End time must be later than start time. Indicate if the constraint represents conditional non-transferability.

